

# RouteRight Power BI Practice Project

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## Initial Assessment and Step 1

Source workbook: `/Users/josealmanza/Desktop/RouteRight_PowerBI_Practice_Dataset.xlsx`

This guide is based on the actual workbook contents.

### A. Dataset Overview

| Sheet / Excel table               | Purpose                                                              | Power BI treatment                  |
|-----------------------------------|----------------------------------------------------------------------|-------------------------------------|
| Read Me                           | Business case, observation window, pilot period, grain, and warnings | Do not import                       |
| Deliveries / DeliveriesTable      | Main delivery-level dataset with 328 records and 34 columns          | Import as the primary fact table    |
| Data Dictionary / DictionaryTable | Definitions and analytical cautions for the fields                   | Do not import; use as documentation |
| Client Reference / ClientsTable   | Eight unique clients and their descriptive attributes                | Import as a client lookup dimension |

#### Actual workbook facts

- Observation window: January 1, 2026 through August 15, 2026.
- Pilot period: May 1, 2026 through August 15, 2026.
- Delivery records: 328.
- Unique `DeliveryID` values: 328.
- Clients: 8.
- Route teams: 4.
- Delivered records: 320.
- Cancelled records: 7.
- In-transit records: 1.
- Retry records: 13.
- Blank `SatisfactionScore` values: 127.
- August records: 21, covering August 1–15.
- All delivery `ClientID` values match the client reference table.
- No duplicate `DeliveryID` values were found.
- No completely blank delivery rows were found.

## Grain

One row represents one delivery record or delivery attempt.

A retry receives a new `DeliveryID` and contains the original delivery's ID in `ParentDeliveryID`.

Therefore:

- `DeliveryID` is unique at the record level.
- Multiple rows may represent one underlying delivery chain.
- The table is not automatically a unique shipment or customer-order table.

## Analytical purpose of the major fields

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### Service outcomes

- `RecordedOnTime` : Reported on-time result using the active definition.
- `StrictOnTime` : On-time result using the stated promise time.
- `DelayMinutes` : Measures lateness or earliness. Negative values mean the delivery was early.
- `DeliveryDurationMinutes` : Operational delivery time.
- `DamagedFlag` : Quality failure indicator.
- `EscalatedFlag` : Customer or operational escalation indicator.
- `SatisfactionScore` : Customer experience outcome.

### Route efficiency

- `PlannedMiles` : Expected route distance.
- `ActualMiles` : Distance actually driven.
- Difference between the two measures route efficiency.
- `DriverInterventions` : Manual intervention or exception-handling workload.
- `CostPerDelivery` : Estimated cost of the delivery record.

### RouteRight exposure

- `InitiativeEligible` : Whether the record qualified for RouteRight.
- `InitiativeUsed` : Whether RouteRight was actually used.
- `RouteTeam` : Operational adoption group.
- `TeamAdoptionDate` : When the team adopted RouteRight.
- `AdoptionLevel` : High, Moderate, or Low adoption classification.

### Segmentation and case mix

- `ClientID`, `ClientName`
- `ClientSize`
- `ServiceTier`

- Region
- RegionType
- Priority
- DeliveryType
- RouteTeam

These fields are important because apparent improvement may simply reflect a different mix of clients, routes, service tiers, or priorities.

## Initial Pre-Pilot versus Pilot snapshot

These are descriptive results, not proof that RouteRight caused the changes. Operational outcome metrics below use delivered records unless noted otherwise.

| Metric                            | Pre-Pilot     | Pilot        | Initial interpretation                               |
|-----------------------------------|---------------|--------------|------------------------------------------------------|
| Delivery records                  | 175           | 153          | Pilot covers fewer calendar days                     |
| Delivered records                 | 170           | 150          | Similar, but Pilot includes partial August           |
| Records per observed day          | 1.46          | 1.43         | Better volume comparison than raw totals             |
| Strict on-time rate               | 51.2%         | 56.7%        | Positive directional signal                          |
| Recorded on-time rate             | 51.2%         | 58.7%        | Not fully comparable because of grace-period records |
| Average delay                     | -0.8 minutes  | -3.6 minutes | Pilot average was earlier                            |
| Average delivery duration         | 102.2 minutes | 98.2 minutes | Positive directional signal                          |
| Mileage variance percentage       | 7.4%          | 2.6%         | Better route efficiency signal                       |
| Average cost per delivered record | \$90.79       | \$88.32      | Positive directional signal, but mix-sensitive       |
| Average driver interventions      | 1.82          | 1.51         | Positive directional signal                          |
| Average satisfaction              | 4.12          | 4.23         | Small positive signal; missingness is important      |
| Damage rate                       | 4.7%          | 4.0%         | Slight improvement                                   |
| Escalation rate                   | 12.9%         | 7.3%         | Positive signal                                      |
| Weather disruption rate           | 4.1%          | 8.0%         | Pilot had more weather disruption                    |

| Metric             | Pre-Pilot | Pilot | Initial interpretation                         |
|--------------------|-----------|-------|------------------------------------------------|
| Contract risk rate | 18.2%     | 12.0% | Positive signal                                |
| Eligible records   | 0         | 118   | RouteRight was not eligible during Pre-Pilot   |
| Used records       | 0         | 74    | RouteRight usage was 62.7% of eligible records |

The initial evidence looks promising, but it must be adjusted for different client and route mix, staggered team adoption, selected non-random RouteRight use, the 15-minute grace-period definition, weather, retries, partial August data, and missing satisfaction responses.

## B. Recommended Power BI Model

Use a star schema.

```

DimDate[Date]
    1
    |
    *
FactDeliveries[DeliveryDate]

DimClient[ClientID]
    1
    |
    *
FactDeliveries[ClientID]

DimRouteTeam[RouteTeam]
    1
    |
    *
FactDeliveries[RouteTeam]

```

### Tables

#### FactDeliveries

Primary fact table based on `DeliveriesTable`. It contains one row per delivery record or attempt.

#### DimClient

Based on `ClientsTable`. It contains one row per `ClientID`.

## DimRouteTeam

Create later from the delivery data using a Power Query Reference query containing:

- RouteTeam
- TeamAdoptionDate
- AdoptionLevel

There is no separate team reference sheet, so this small dimension must be derived from DeliveriesTable .

## DimDate

Create later as a DAX date table.

## Measures table

Optionally create a disconnected table called Measures . It has no relationships and is only used to organize DAX measures.

## Relationships

| From                    | To                           | Cardinality | Direction |
|-------------------------|------------------------------|-------------|-----------|
| DimDate[Date]           | FactDeliveries[DeliveryDate] | One-to-many | Single    |
| DimClient[ClientID]     | FactDeliveries[ClientID]     | One-to-many | Single    |
| DimRouteTeam[RouteTeam] | FactDeliveries[RouteTeam]    | One-to-many | Single    |

Do not create an active relationship between FactDeliveries[ParentDeliveryID] and FactDeliveries[DeliveryID] . That self-relationship can create confusing filter behavior and duplicate-counting risks. Analyze retries with calculated columns and measures instead.

## C. Recommended KPIs

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1. Total delivery records.
2. Delivered delivery records.
3. Completion rate.
4. Records per observed day.
5. Strict on-time delivery rate.
6. Average delay minutes.
7. Mileage variance percentage.
8. Average cost per delivered record.
9. Average driver interventions.
10. Average satisfaction with satisfaction response rate.
11. Damage and escalation rates.
12. RouteRight eligibility and usage rate.

Additional risk indicators include contract risk rate, weather disruption rate, retry rate, retry records, cancelled records, and in-transit records.

## D. Recommended Dashboard Pages

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### Page 1 — Executive Overview

Purpose: Give management a concise view of performance and decision evidence.

Recommended content:

- KPI cards for strict on-time rate, average delay, mileage variance, cost, satisfaction, usage, and risk.
- Pre-Pilot versus Pilot comparison matrix.
- Monthly trend for strict on-time performance.
- Monthly trend for average delay.
- A note explaining that Pilot results are observational, not causal.
- A decision-support indicator showing Continue, Revise, or End only after thresholds are agreed.

### Page 2 — RouteRight Usage and Adoption

Purpose: Analyze whether RouteRight was adopted and whether used records perform differently.

Recommended content:

- Eligible records.
- Used records.
- Usage rate among eligible records.
- Usage rate by route team.
- RouteRight used versus not used comparison.
- Adoption date timeline.
- Performance by adoption level.
- Performance by days since adoption.

### Page 3 — Operational Performance

Purpose: Analyze route efficiency and delivery execution.

Recommended content:

- Strict on-time rate by month.
- Recorded versus strict on-time rate.
- Average delay by month.
- Average delivery duration.
- Planned miles versus actual miles.
- Mileage variance percentage.
- Average driver interventions.

- Average cost per delivered record.

## Page 4 — Client and Route Analysis

Purpose: Determine whether results are concentrated in certain segments.

Recommended content:

- On-time rate by client.
- Cost by client.
- Performance by client size.
- Performance by service tier.
- Performance by region and region type.
- Performance by route team.
- Matrix containing delivery volume, strict on-time rate, cost, delay, and risk.

## Page 5 — Risk and Exceptions

Purpose: Identify operational or customer risks.

Recommended content:

- Damage rate.
- Escalation rate.
- Contract risk rate.
- Weather disruption rate.
- Satisfaction response rate.
- Low satisfaction deliveries.
- Retry records.
- Cancelled and in-transit records.
- Drill-through detail table for individual exceptions.

## E. Important Data Issues

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### 1. August is incomplete

August contains only August 1–15.

August has 21 delivery records, 20 delivered records, and `DaysObservedInMonth = 15`.

Do not compare raw August volume with a full month.

A simple normalized August record estimate is:

```
August normalized records
= August raw records ÷ 15 × 31
```

For this dataset:

$21 \div 15 \times 31 = 43.4$  estimated records

For delivered records:

$20 \div 15 \times 31 = 41.3$  estimated delivered records

These are estimates, not actual observations. Keep both raw August values and clearly labeled normalized estimates. Do not modify the source rows.

Rates and averages are less affected by the partial month than totals, but August results may still reflect a different mix or short-term conditions.

## 2. Recorded on-time performance is not fully comparable

The dataset contains 294 records using `By Promise Time` and 34 Pilot records using `15-Minute Grace`.

All 34 grace-period records are in the Pilot and were RouteRight-used records.

Use `StrictOnTime` as the primary Pre-Pilot versus Pilot metric. Use `RecordedOnTime` as a secondary metric, with `OnTimeDefinition` visible or filtered.

## 3. Satisfaction is missing non-randomly

There are 127 blank satisfaction scores.

Among delivered records:

- 119 of 320 satisfaction scores are blank.
- 75.8% of escalated records have missing satisfaction.
- 57.1% of damaged records have missing satisfaction.
- 52.6% of weather-disrupted records have missing satisfaction.

Do not replace missing satisfaction with zero. Show average satisfaction, satisfaction response count, satisfaction response rate, and missing satisfaction count.

## 4. Status affects denominators

The data contains 320 Delivered, 7 Cancelled, and 1 In Transit.

For on-time, delay, duration, and satisfaction KPIs, use `Status = Delivered`. Do not delete cancelled or in-transit records. Analyze them separately.

## 5. Retries

There are 13 retry records with a nonblank `ParentDeliveryID`.

Count retries separately for operational workload, attempt volume, cost, retry rate, and RouteRight behavior on retry attempts.

For a unique delivery-chain count, create:

```
DeliveryChainID =  
if ParentDeliveryID is blank  
then DeliveryID  
else ParentDeliveryID
```

Then use:

```
DISTINCTCOUNT(FactDeliveries[DeliveryChainID])
```

Do not delete retries automatically.

## 6. Route teams adopted at different times

| Route team | Adoption date | Adoption level |
|------------|---------------|----------------|
| Team Atlas | May 1, 2026   | High           |
| Team Brook | May 15, 2026  | High           |
| Team Cedar | June 1, 2026  | Moderate       |
| Team Delta | July 1, 2026  | Low            |

Team performance is not directly causal because teams handle different routes and adopted at different times.

## 7. July weather should not be deleted

The Read Me and Data Dictionary identify July weather disruptions as legitimate extremes. Keep them and use `WeatherDisruptionFlag` to stratify or explain results.

## 8. Contract risk is a screening flag

`ContractRiskFlag` should help identify deliveries requiring management attention. It should not automatically be interpreted as a final contract decision.

## Field-use table

| Column                        | Use in Power BI? | Why?                                                                | Role                                                     |
|-------------------------------|------------------|---------------------------------------------------------------------|----------------------------------------------------------|
| <code>DeliveryID</code>       | Yes              | Unique delivery-record key                                          | Key, drill-through, data-quality check                   |
| <code>ParentDeliveryID</code> | Yes              | Identifies retry records                                            | Retry flag, chain logic, data-quality check              |
| <code>DeliveryDate</code>     | Yes              | Supports time analysis                                              | Date relationship, axis, slicer                          |
| Month                         | Limited          | Source month label can be checked against <code>DeliveryDate</code> | Data-quality check; use <code>DimDate</code> for visuals |

| Column                  | Use in Power BI? | Why?                                       | Role                                          |
|-------------------------|------------------|--------------------------------------------|-----------------------------------------------|
| Period                  | Yes              | Separates Pre-Pilot and Pilot              | Category, slicer, comparison dimension        |
| DaysObservedInMonth     | Yes              | Supports August normalization              | Normalization measure; use MAX , never SUM    |
| ClientID                | Yes              | Connects deliveries to client reference    | Relationship key                              |
| ClientName              | Yes              | Identifies the client                      | Dimension/category; preferably from DimClient |
| ClientSize              | Yes              | Controls for client mix                    | Dimension/category, slicer                    |
| ServiceTier             | Yes              | Controls for service complexity            | Dimension/category, slicer                    |
| Region                  | Yes              | Supports geographic analysis               | Dimension/category, slicer                    |
| RegionType              | Yes              | Urban, suburban, or rural comparison       | Dimension/category, slicer                    |
| RouteTeam               | Yes              | Measures adoption and team performance     | Dimension/category, slicer                    |
| TeamAdoptionDate        | Yes              | Determines exposure timing                 | Date analysis, adoption dimension             |
| AdoptionLevel           | Yes              | Groups team adoption maturity              | Category, slicer, comparison                  |
| Priority                | Yes              | Major case-mix factor                      | Category, slicer, adjustment dimension        |
| DeliveryType            | Yes              | Identifies delivery complexity/type        | Category, slicer                              |
| Status                  | Yes              | Controls KPI denominators                  | Filter, slicer, measure logic                 |
| InitiativeEligible      | Yes              | Identifies records eligible for RouteRight | Filter and measure logic                      |
| InitiativeUsed          | Yes              | Identifies actual RouteRight exposure      | Filter and measure logic                      |
| PlannedMiles            | Yes              | Establishes expected route distance        | Numerical metric                              |
| ActualMiles             | Yes              | Measures actual route distance             | Numerical metric                              |
| DelayMinutes            | Yes              | Measures early or late performance         | Numerical measure                             |
| DeliveryDurationMinutes | Yes              | Measures execution time                    | Numerical measure                             |
| RecordedOnTime          | Yes, cautiously  | Shows reported on-time result              | Measure; definition-sensitive                 |
| StrictOnTime            | Yes              | Provides comparable on-time result         | Primary KPI measure                           |

| Column                | Use in Power BI? | Why?                                    | Role                               |
|-----------------------|------------------|-----------------------------------------|------------------------------------|
| OnTimeDefinition      | Yes, cautiously  | Explains measurement differences        | Filter and data-quality check      |
| DamagedFlag           | Yes              | Measures delivery damage                | Risk measure and filter            |
| EscalatedFlag         | Yes              | Measures escalations                    | Risk measure and filter            |
| SatisfactionScore     | Yes              | Measures customer experience            | Average measure; missingness check |
| CostPerDelivery       | Yes              | Measures delivery economics             | Average, total, and cost measures  |
| DriverInterventions   | Yes              | Measures manual routing workload        | Total and average measures         |
| WeatherDisruptionFlag | Yes              | Explains unusual operational conditions | Risk measure, filter, stratifier   |
| ContractRiskFlag      | Yes              | Screens for service-risk cases          | Risk measure and filter            |

## Business questions the dashboard should answer

1. Did delivery reliability improve during the Pilot?
2. Did strict on-time performance improve?
3. Did average delay and delivery duration improve?
4. Did actual miles move closer to planned miles?
5. Did cost per delivered record improve?
6. Did driver interventions decrease?
7. Did customer satisfaction improve, and how complete are the responses?
8. Did damage, escalations, or contract risk increase or decrease?
9. What percentage of eligible deliveries actually used RouteRight?
10. Which route teams adopted and used RouteRight most consistently?
11. Are benefits concentrated in particular clients, service tiers, regions, or priorities?
12. Are improvements still visible after reviewing weather and retry records?

The data is observational. It does not contain a randomized control group. The dashboard can show evidence consistent with RouteRight helping, but it cannot prove causation.

## Complete Build Plan

1. Import only the correct Excel tables.
2. Validate row counts and column names.
3. Set Power Query data types.
4. Handle blank values without destroying information.
5. Create retry fields.
6. Validate client-reference consistency.

7. Create `DimRouteTeam`.
8. Apply the queries.
9. Create `DimDate`.
10. Build relationships.
11. Create DAX measures.
12. Validate measures against the source.
13. Build the Executive Overview.
14. Build RouteRight adoption analysis.
15. Build operational analysis.
16. Build client and route analysis.
17. Build risk and exception analysis.
18. Add slicers, tooltips, drill-through, navigation, and formatting.
19. Perform Pre-Pilot versus Pilot interpretation.
20. Complete the final recommendation and quality-control checklist.

## STEP 1 — Import the correct Excel tables

### What we are doing

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We are connecting Power BI to the workbook and selecting only:

- `DeliveriesTable`
- `ClientsTable`

We will not load the Read Me or Data Dictionary into the analytical model.

### Where to click

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1. Open Power BI Desktop.
2. Select **Home**.
3. Select **Get data**.
4. Choose **Excel workbook**.
5. Browse to `/Users/josealmanza/Desktop/RouteRight_PowerBI_Practice_Dataset.xlsx`.
6. Select the file.
7. Click **Open**.

### What to select

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In the Navigator window, select:

- `DeliveriesTable`
- `ClientsTable`

Do not select:

- Read Me
- DictionaryTable
- Data Dictionary

If Power BI only displays worksheets rather than named tables, select:

- Deliveries
- Client Reference

The named Excel tables are preferred because their ranges are explicitly defined:

- DeliveriesTable: A1:AH329
- ClientsTable: A1:F9

## Continue to Power Query

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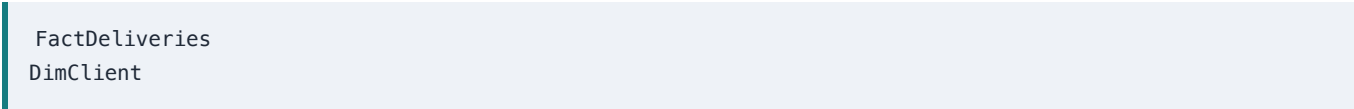
1. Click **Transform Data**.
2. Wait for Power Query Editor to open.
3. In the Queries pane, right-click `DeliveriesTable`.
4. Select **Rename**.
5. Rename it `FactDeliveries`.
6. Right-click `ClientsTable`.
7. Rename it `DimClient`.

Do not click **Close & Apply** yet.

## Expected result

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You should see two queries:



```
FactDeliveries
DimClient
```

Expected row and column counts:

- `FactDeliveries`: 328 rows, 34 columns.
- `DimClient`: 8 rows, 6 columns.

If those counts do not match, stop before making any transformations and record what Power BI shows.

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End of the first guided lesson. Continue only after the import and row counts are confirmed.